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Capstone Project Phase A

Comparative Study of Single Marker Test and Single Interval Mapping for QTL Detection Algorithms in a Single Backcross Family

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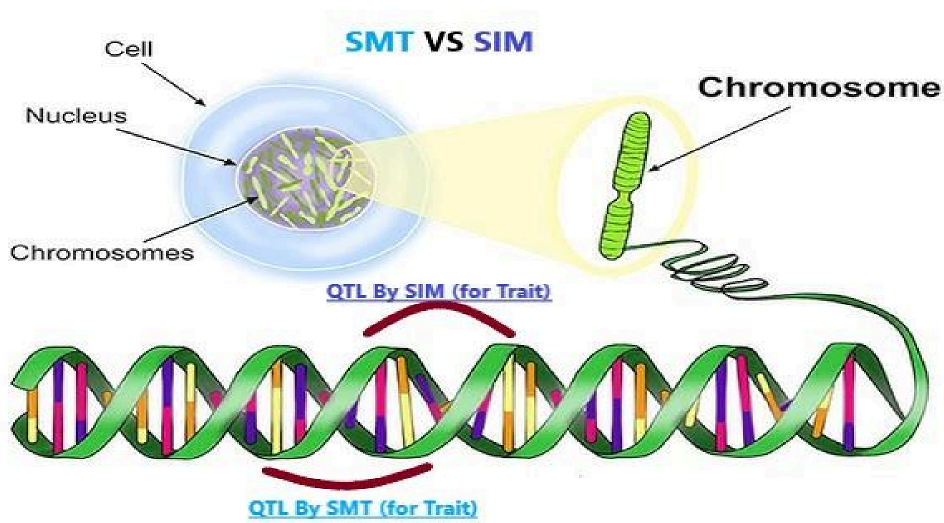

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https://github.com/Tamert98/QTL-Mapping-Project

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Abstract

Quantitative Trait Loci (QTL) mapping is a fundamental tool in genetic research, enabling the identification of genomic regions associated with quantitative traits. This study conducts a comparative analysis of two widely employed QTL detection methodologies—Single Marker Test (SMT) and Single Interval Mapping (SIM) [3] in the context of a single backcross family. By leveraging the availability of precise physical positions of genetic markers, the research evaluates the performance of these methods based on accuracy, statistical power, and computational efficiency. Using both real genetic data in Variant Call Format (VCF) and simulated datasets, the study provides insights into the strengths and limitations of SMT and SIM. It highlights the impact of marker density, trait heritability, and data noise on method performance, offering actionable recommendations for selecting optimal approaches in genetic studies.

1. Introduction

Quantitative traits, such as height, yield, or disease resistance, are often controlled by multiple genetic loci known as **Quantitative Trait Loci (QTLs)**. Identifying and mapping these loci is essential for understanding the genetic architecture of complex traits, particularly in agricultural and biological research. QTL mapping involves statistical methods that link phenotypic traits with molecular markers to locate genomic regions influencing these traits. This information is crucial for applications such as marker-assisted selection (MAS), gene discovery, and breeding programs.

Two widely used methods for QTL detection are the **Single Marker Test (SMT)** and **Single Interval Mapping (SIM)**:

- **Single Marker Test (SMT)** identifies QTLs by testing associations between individual genetic markers and trait values. While straightforward and computationally efficient, SMT may fail to account for QTLs located between markers.
- **Single Interval Mapping (SIM)** improves QTL localization by using nearby markers and likelihood-based statistical models to estimate QTL positions within known specific sections of the genome. This approach provides higher accuracy but requires more computational resources.

This study focuses on comparing SMT and SIM in the context of a **single backcross family**. The primary objective is to evaluate these two methods across three key aspects:

1. **Accuracy**: How well a method identifies the true location of QTLs.
2. **Statistical Power**: The ability of a method to detect QTLs when they truly exist (i.e., avoiding false negatives).
3. **Computational Efficiency**: How quickly and resource-effectively a method processes data to produce results.

Given the availability of high-resolution marker positions, this study aims to assess how well each method identifies QTLs under ideal and practical conditions. The use of precise physical marker positions eliminates positional ambiguity, allowing for robust comparisons between the methods.

The research utilizes real genetic data stored in **Variant Call Format (.vcf)**, a standard file format for genomic variation. Additionally, **simulated data** will be used to evaluate the performance of SMT and SIM under controlled conditions, Simulated datasets lead to precise testing of accuracy, statistical power, and computational efficiency by providing known QTL positions and controlled variations, such as marker density, sample size, and data noise.

The simulations and analyses will be conducted using the **R Studio workspace**, a user-friendly environment for R programming. Within R Studio, the **qtl library [20]** will be employed to facilitate QTL analysis and data simulation. The **qtl library** offers powerful tools for generating simulated datasets that mimic real genetic data, defining genetic maps, and analyzing QTLs through methods like Single Marker Test (SMT) and Single Interval Mapping (SIM).

By leveraging high-resolution marker positions, this study aims to provide new insights into the performance of SMT and SIM for QTL detection in backcross populations. The findings will contribute to understanding the practical trade-offs between accuracy, computational efficiency, and statistical power, guiding researchers in selecting optimal mapping methods for genetic studies.

2. Background and related work

Quantitative Trait Loci (QTL) mapping has been a cornerstone in identifying genomic regions associated with quantitative traits. Two widely used approaches in this domain are **Single Marker Analysis (SMA)** and **Interval Mapping (IM)**.

Single Marker Analysis (SMA) evaluates individual markers for their association with the phenotype, relying on straightforward statistical techniques such as regression or ANOVA. While computationally efficient, SMA often faces limitations, including confounding between QTL effect size and recombination frequency, as well as difficulties in resolving closely linked QTLs. This can lead to biased estimates and lower power in complex genetic architectures [13].

In contrast, **Interval Mapping (IM)**, first introduced by Lander and Botstein (1989), builds on the principles of linkage analysis by utilizing intervals flanked by two markers. IM applies maximum likelihood estimation to identify the probable position and effect of a QTL within a chromosomal region. This approach accounts for missing genotypes and marker linkage, offering higher precision and statistical power compared to SMA [11].

Both methodologies have their respective strengths and limitations. SMA is well-suited for preliminary analyses with sparse marker data, whereas IM excels in dense marker datasets, providing finer resolution for QTL detection. These approaches have laid the foundation for advanced techniques like Composite Interval Mapping and Genome-Wide Association Studies, further enhancing our ability to dissect complex traits.

2.1. The Model

We consider a single backcross family: two parents (I_{00} and I_{01}, from pure line and hybrid, respectively) and a set of children I_i, I=1,...,n. Genotype g_i of the organism i is defined by states of M+Q biallelic loci: M markers and Q QTLs. Physical position of locus m is the same for all organisms and defined by pair (chr_m, $coor_m$) where chr is the chromosome name (from the set of CHR chromosomes) and $coor$ is integer number characterizing nucleotide position of the genetic variation corresponding to this locus. Number and position of QTLs is unobservable while positions of all M markers are given. Genotype g_{im} of organism i in locus m is defined by a set of two alleles (g_{im0}, g_{im1}) inherited from the mother and from the father, respectively (for all children individuals, from I_{00} and I_{01}, respectively). Each allele in locus m has only two possible states defined by A_m and a_m. In classical design, all loci of I_{00} are expected to be fixed while loci of I_{01} are expected to be hybrid: g_{00m0}=g_{00m1}=g_{01m0}=A_m, g_{01m1}=a_m. Allele states in QTL loci are unobserved. Allele states in M marker loci of I_{00} and I_{01} and n children loci are observed with some amount of errors and missing data. In locus m, each child inherits one allele of g_{00m0} and g_{00m1} (expected to be A_m) and one allele of g_{01m0} and g_{01m1} (expected to be A_m or a_m with equal probability 0.5).

Inheritance of alleles in different children or on different chromosomes is considered to be independent, while inheritance within a single chromosome is defined by "Haldane model of recombination". According to this model, recombination events of different intervals are independent. Each interval between two loci $m1$ and $m2$ on the same chromosome is characterized by recombination rate r_{m1m2}. If g_{im1}=g_{01m1h} then g_{im21}=g_{01m2h} or $g_{01m2(1-h)}$ with probabilities (1-r_{m1m2}) and r_{m1m2}, respectively; h=0, 1. Recombination rates are defined by marker positions and can be estimated based on children genotypes (see below).

Quantitative trait T_i of individual i is associated with unobserved genotype in QTL loci. We use Gaussian model of trait: T_i is calculated as Gaussian random number with mean μ_i and variance σ_i^2 defined by the genotype in QTL loci:

$\mu_i=\mu+\sum_q \mu_q(1\{g_{iq0}=a_q\}+1\{g_{iq1}=a_q\})$,

$\sigma_i=\sigma+\sum_q \sigma_q(1\{g_{iq0}=a_q\}+1\{g_{iq1}=a_q\})$.

Trait values T_i are observed (possibly, with some errors and some missing data), while parameters μ, μ_q, σ and σ_q are unknown.

In terms of this model QTL mapping problem is formulated as following: based on given genotypes g_{im} of n individuals in M markers with known physical positions (chr_m, $coor_m$) and trait values T_i to estimate the number Q of QTLs, their positions (chr_q, $coor_q$) and effects on trait (μ_q and σ_q). In general, this problem is quite difficult. We start to deal with the simple case Q=1 and μ_q or σ_q =0. If we have enough time, we also will consider some more complicated cases (e.g., with Q=2 or 3).

2.2 Estimation of recombination rates, genetic map

In real situations only traits and marker genotypes of the individuals are observed. To take putative recombination events into account one can introduce unobserved recombination frequencies into the model. In simple cases it can be supposed that recombination events along the chromosome are described by the **Haldane model**, which assumes that recombination rates r_{m1m2} between two loci *m1* and *m2* are independent and follow a **Poisson process**.

This foundational concept was introduced in Haldane's seminal work on the mathematical theory of chromosome behavior during meiosis [12].

In this case all recombination rates can be defined from the position of all loci on one-dimensional genetic map: Let loci *m1* and *m2* are situated on the same chromosome in positions x_{m1} and x_{m2} respectively (scored in centiMorgans). According to Haldane model,

$$\boldsymbol{r_{m1m2}} = 0.5\,(1 - exp\{-\ 0.02\ *\ \boldsymbol{d_{m1m2}}\})$$

$$\boldsymbol{d_{m1m2}} = x_{m2} - x_{m1}$$

i.e., r_{m1m2} =0.01 for markers situated on distance of one centiMorgan and monotonically increasing to 0.5 (free recombination) with genetic distance goes to infinity.

In practice, genetic maps are unobserved. For genetic markers genetic map position can be reconstructed by calculating recombination rates and converting these rates to genetic distances by formulas (based on maximal likelihood estimations for these parameters):

2.3 Estimation of QTL position

2.3.1. Single Marker Test (SMT)

The **Single Marker Test (SMT)** is a straightforward statistical method used in Quantitative Trait Loci (QTL) analysis to examine the association between individual genetic markers and a quantitative trait. It is widely appreciated for its computational efficiency and simplicity, making it a practical starting point in QTL mapping.[8]

2.3.1.1. Key Concepts:

1. **Genotypic States**: For a biallelic marker, the genotypes can be represented as AA, Aa, or aa, where A and a are alleles at the marker locus.
2. **Trait Distribution**: The quantitative trait Ti is modeled as a continuous variable, often assumed to follow a normal distribution.

In SMT, each marker is analyzed independently by testing whether there is a statistically significant difference in the quantitative trait values associated with the different genotypes at the marker locus. This method assumes the following linear regression model [5] for trait Ti for individual i:

$$Ti = \mu + \beta\ Gim + \epsilon i$$

Where:

- Ti: Trait value for individual i.
- μ: Overall mean of the trait.
- Gim: Genotype of individual i at marker m (coded as 0, 1, or 2 for additive effects in biallelic markers), for example:
  - Gim = 0: Alternative allele (aa).
  - Gim=1: Heterozygous (Aa).
  - Gim=2: Main allele (AA).
- β: Marker effect, indicating the strength and direction of association.
- ϵi: Residual error, assumed $\epsilon i \sim N(0, \sigma 2)$.

The hypothesis tested in SMT is:

- Null Hypothesis (H0): β = 0 (no association between the marker and the trait).
- Alternative Hypothesis (H1): β ≠ 0(association exists between the marker and the trait).[1]

The test statistic is typically computed using an analysis of variance - F-test [21] or a t-test for the regression coefficient β, depending on the coding of Gim. The corresponding p-value determines whether the null hypothesis H0 can be rejected.

2.3.1.2. Implementation:

1. **Marker-Trait Association**: For each marker, calculate:
    - The regression coefficient β^:

$$\hat{\beta} = \frac{\sum_i (G_{im} - \bar{G}_m)(T_i - \bar{T})}{\sum_i (G_{im} - \bar{G}_m)^2}$$

    - Gim: Mean the genotype of individual i.
    - G¯m: Mean the the average of the genotypes at the marker m.
    - T¯: Mean the average of the trait in the population.
    - Ti: Mean the value of the trait of individual i.

2. **Test Statistic**: Compute the test statistic:

$$F = \frac{SS_{\text{regression}}/1}{SS_{\text{error}}/(n-2)}$$

Where:

- SSregression: Sum of squares of variance explained by the marker.
- SSerror: Sum of squares of errors (noise).
- n: Number of individuals.
    - n-2: Degrees of freedom of the unexplained variance, since we are estimating two parameters: μ (the mean) and β (the regression coefficient).

2.3.1.3. Advantages:

1. **Computational Efficiency**: SMT processes each marker individually, making it less computationally intensive compared to multi-marker approaches.
2. **Ease of Implementation**: Simple statistical tests and straightforward interpretation make SMT accessible for researchers.
3. **Scalability**: Suitable for datasets with a large number of markers.

2.3.1.4. Limitations:

1. **Lack of Resolution**: SMT only evaluates markers individually and does not consider the genetic linkage or interactions between markers, which can reduce its power to detect QTLs in regions with sparse marker coverage or linked QTLs. [3]
2. **Power**: Its reliance on single-marker associations limits its power in detecting QTLs influenced by multiple linked markers.
3. **Multiple Testing**: Analyzing each marker independently increases the risk of false positives. Corrections (e.g., Bonferroni) are needed to control for multiple testing.[1]

2.3.1.5. Conclusion:

While SMT offers a simple and efficient approach to QTL mapping, its limitations in resolution and statistical power make it more suitable as an initial exploratory tool rather than a comprehensive mapping method. For more robust QTL detection, methods like **Single Interval Mapping (SIM)** [14] provide significant improvements by integrating information from neighboring markers.

2.3.2. Single Interval Mapping (SIM)

2.3.2.1. Key Concepts

Single Interval Mapping (SIM) [3] builds on the limitations of SMT by considering the likelihood of QTL presence within intervals between markers. This method incorporates information from flanking markers, providing a more refined estimation of QTL positions. SIM uses maximum likelihood estimation (MLE) to calculate LOD (logarithm of odds) scores for each interval, improving both power and resolution. The LOD score is defined as:

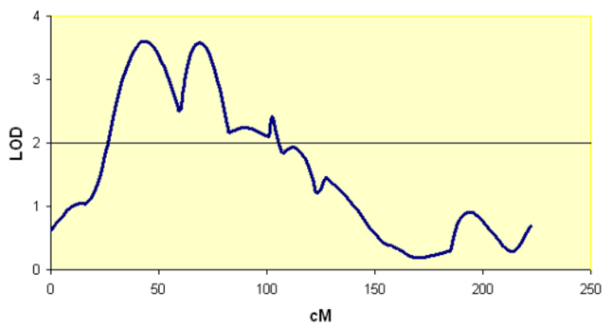
$$LOD = ln(\frac{L}{L_0})$$

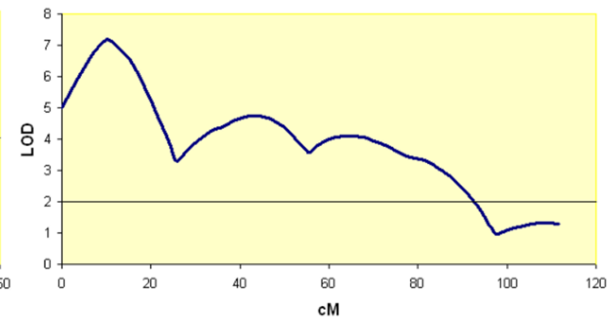
where :

L : The likelihood of the data given a QTL is present in the interval (see below).

L_0 : The likelihood of the data given no QTL is present (see below).

By focusing on intervals, SIM reduces false positives and improves localization precision. In case of no QTL, value X^2=2 LOD should have χ^2 distribution with degrees of freedom equal to number of fixed parameters (in case of single QTL q, we need 4 parameters x_q, μ, μ_q and σ, but only additive effect is fixed, hence d.f.=1). Hence, position x_q where LOD is maximal can be considered as maximal likelihood estimation of QTL position on the genetic map. Statistical p-value of QTL presence can be calculated via X^2-test or in permutation tests (see below).





2.3.2.2 Likelihood

For a set of n offspring we can suppose that trait values and recombination events are independent. In classical design, all loci of I_{00} are expected to be fixed while loci of I_{01} are expected to be hybrid: $g_{00m0}=g_{00m1}=g_{01m0}=A_m$, $g_{01m1}=a_m$. In this case, all n individuals at locus m (marker or QTL) inherit from I_{00} allele A_m, but from I_{01} in allele A_m or a_m with probabilities 0.5. For interval $m1$-q-$m2$ probabilities for children haplotypes (allele combinations) inherited from I_{01} are:

$\Pr(A_{m1}A_qA_{m2})=0.5(1-r_{m1q})(1-r_{qm2})$

$\Pr(A_{m1}A_qa_{m2})=0.5(1-r_{m1q})r_{qm2}$

$\Pr(A_{m1}a_qA_{m2})=0.5\ r_{m1q}\ r_{qm2}$

$\Pr(A_{m1}a_qa_{m2})=0.5\ r_{m1q}\ (1-r_{qm2})$

$\Pr(a_{m1}A_qA_{m2})=0.5\ r_{m1q}\ (1-r_{qm2})$

$\Pr(a_{m1}A_qa_{m2})=0.5\ r_{m1q}\ r_{qm2}$

$\Pr(a_{m1}a_qA_{m2})=0.5(1-r_{m1q})r_{qm2}$

$\Pr(a_{m1}a_qa_{m2})=0.5(1-r_{m1q})(1-r_{qm2})$

Based on these values one can calculate $Pr(h|x_q, g_{m1}, g_{m2})$.

Likelihood for single QTL case is calculated by:

$$L(T|\ x_q, G_{m1}, G_{m2}, \mu, \mu_q, \sigma) = \prod_{i=1}^{n} L(T_i|x_q, g_{im1}, g_{im2}, \mu, \mu_q, \sigma)$$

$$L(T_i|x_q, g_{im1}, g_{im2}, \mu, \mu_q, \sigma) = \sum_h Pr(h|x_q, g_{im1}, g_{im2}) Pr(T_i|h, \mu, \mu_q, \sigma)$$

$$Pr(T_i|h, \mu, \mu_q, \sigma) = \frac{1}{\sqrt{2\pi\sigma}} exp(- \frac{(T_i - T(\mu(h)))^2}{2\sigma^2})$$

$\mu(h)=\mu$ or $\mu+\mu_q$ for h with A_q and a_q respectively.

$L=\max L(T|x_q,G_{m1},G_{m2},\mu,\mu_q,\sigma)$

$L_0=\max L(T|x_q,G_{m1},G_{m2},\mu,\mu_q=0,\sigma)=\max L(T|\mu,\mu_q=0,\sigma)=\max_{\mu,\sigma} L(T|\mu,\sigma)$

2.3.2.3. Permutation test

Permutation tests are a crucial statistical tool used in Single Interval Mapping (SIM) to assess the significance of QTL presence and reduce false discoveries. The test works by generating a null distribution of the test statistic (e.g., LOD score) under the hypothesis that no QTL is present. This is achieved by permuting the trait values (***T_i***) across individuals, breaking the relationship between the trait and genotypes while preserving the original genetic structure.

For each permutation, the SIM procedure is repeated to calculate the maximum LOD score across the genome. This process is performed multiple times (e.g., 1000 permutations) to construct the null distribution of maximum LOD scores. The significance threshold is then determined based on this distribution, often selecting the LOD value corresponding to a desired significance level (e.g., 5% or p=0.05).

This threshold is used to identify significant QTLs in the observed data. If the observed maximum LOD score exceeds the threshold, the QTL is considered statistically significant. Permutation tests are especially valuable in QTL mapping as they account for genome-wide multiple testing, ensuring a robust control of false positives. By leveraging this approach, SIM provides a reliable framework for identifying and localizing QTLs while maintaining statistical rigor. [5]

2.3.2.4. Resampling

Resampling methods are powerful tools in statistical analysis that involve generating new datasets from the original data to evaluate the reliability and stability of statistical results. In QTL mapping, resampling allows researchers to assess the robustness of detected QTLs and estimate the uncertainty associated with their positions and effects. By repeatedly analyzing resampled datasets, resampling provides insights into the consistency of results under different sampling scenarios. This approach helps mitigate the effects of sampling variability, improves the reliability of conclusions, and facilitates better interpretation of genetic influences on quantitative traits.[9]

Bootstrap :

The bootstrap method is a widely used resampling technique in QTL mapping, designed to assess the precision and reliability of statistical estimates. This method involves generating multiple resampled datasets by randomly sampling individuals **with replacement** from the original dataset, maintaining the same sample size. Each resampled dataset is independently analyzed to detect QTLs and estimate their parameters, such as position and effect size. By evaluating the distribution of QTL estimates across all bootstrap replicates, confidence intervals for these parameters can be constructed. For example, a 95% confidence interval for a QTL position is derived from the range encompassing 95% of the bootstrap replicates. The bootstrap method is especially valuable in cases with small sample sizes or missing data, as it provides empirical measures of uncertainty and enhances the robustness of QTL mapping results.[9]

Jackknife:

The jackknife method is another resampling technique that systematically evaluates the sensitivity of statistical results to individual observations. Unlike the bootstrap, the jackknife method generates **n** distinct subsets by sequentially omitting one observation at a time from the original dataset (where **n** is the total number of individuals). Each subset is analyzed separately to assess the impact of removing a specific data point on the overall results. This technique is particularly useful for identifying influential observations that may disproportionately affect QTL detection. The jackknife method also provides estimates of variability and bias in QTL parameters, making it a complementary approach to the bootstrap for improving the robustness and interpretability of QTL mapping analyses.[19]

2.3.2.5. Implementation

The implementation of SIM involves the following steps:

1. Calculate L_0=max$_{\mu,\sigma}$ $L(T|\mu,\sigma)$ and (μ_0,σ_0)=argmax$_{\mu,\sigma}$ $L(T|\mu,\sigma)$ by gradient descent method
2. For each chromosome *chr*:
   2.1. Select the set M_{chr} of markers presented on *chr*
   2.2. For each marker *m* from M_{chr} calculate its position x_m on the genetic map using the Haldane model.
   2.3. For each possible candidate position x_q on genetic map for chromosome *chr*
      2.3.1 For each offspring individual *i*
         2.3.1.1 Select one or two closest flanking markers m_1 and m_2 for x_q with clear genotypes g_{im1} and g_{im2}.

2.3.1.2 Define $L_i(x_q)=(T_i|x_q,g_{im1},g_{im2},\mu,\mu_q,\sigma)$

2.3.2 Calculate $L(x_q)$ and $(\mu,\mu_q,\sigma)(x_q)$ by gradient descent method

2.4. Calculate $LOD(x_q)$

2.5. Select x_q with maximal $LOD(x_q)$ and extract corresponding parameters μ,μ_q,σ

2.6 Estimate p-value and confidence intervals for the parameters based on χ^2 test

2.7 Estimate p-value in permutation test

2.8 Estimate confidence intervals for parameters based on sampling of individuals and markers

3. Drew graphs and save reports

2.3.2.6. Advantages

1. Accounts for flanking markers, providing better power and resolution compared to SMT.
2. Reduces false positives by leveraging interval-specific likelihoods.
3. Offers precise localization of QTLs.
4. Handles continuous traits effectively with fewer assumptions.
5. Can accommodate missing genotype data through MLE.

2.3.2.7. Limitations

1. Computationally intensive due to iterative MLE over all intervals.
2. Requires accurate marker positions and recombination rates.
3. Assumes independence of recombination events, which may not hold universally.
4. Performs poorly with sparsely distributed markers.
5. Limited to single-QTL models and does not account for QTL interactions.
6. Sensitive to errors in genotype and phenotype data.

2.3.2.8. Conclusion

Single Interval Mapping (SIM) provides a robust and refined approach to QTL identification by integrating information from flanking markers. While computationally demanding, its precision and ability to reduce false positives make it an essential tool in genetic analysis. Future extensions could focus on addressing multiple QTLs and incorporating more complex models for marker interactions and missing data.

2.4 Comparison of the methods

Previous studies comparing **SMT (Single Marker Test)** and **SIM (Single Interval Mapping)** have highlighted key differences in their performance and utility in QTL mapping. While SMT is computationally simpler and performs well in datasets with high marker density, it tends to lose power in identifying QTLs when markers are unevenly spaced or when the effects of flanking markers are significant. In contrast, SIM improves localization accuracy by incorporating information from flanking markers, making it better suited for datasets with uneven marker spacing or lower marker density. [3]

2.4.1 Power and Sensitivity:

Power, defined as the ability of a method to detect true QTLs (true positives), is generally higher in SIM than in SMT, particularly when QTLs are positioned in intervals between markers. SMT's reliance on individual marker analysis may result in missed QTLs (false negatives) if the markers do not directly capture the QTL. SIM's likelihood-based framework increases its sensitivity by leveraging flanking marker information, allowing it to identify QTLs more reliably even in the presence of recombination. [7]

2.4.2 Accuracy and Predictive Performance:

Accuracy in QTL mapping is influenced by both the true positive rate (sensitivity) and the false positive rate (specificity). SIM tends to have higher accuracy than SMT because it evaluates the likelihood of QTL presence within intervals, reducing false positives. Receiver Operating Characteristic (ROC) curves can illustrate the trade-off between sensitivity and specificity, and studies have shown that SIM generally exhibits a higher area under the ROC curve (AUC), reflecting better overall performance.[21]

2.4.3 False Discovery Rate (FDR):

FDR [1] , the proportion of false positives among all significant findings, is a critical metric in QTL mapping. SMT, by analyzing markers individually, often yields higher FDR due to its susceptibility to noise and spurious associations in datasets with large numbers of markers. SIM reduces FDR by incorporating flanking marker data, thereby increasing confidence in detected QTLs.

2.4.4 Robustness and Data Quality Dependence:

Both SMT and SIM are heavily dependent on the quality of genetic data, including accurate marker positions, reliable phenotypic measurements, and appropriate handling of missing data. However, SIM is generally more robust to imperfections in marker density because it integrates information across intervals. [3]

2.4.5 Practical Considerations:

In practical applications [14] , the choice between SMT and SIM depends on the specific dataset and research objectives. SMT may be preferred for exploratory analyses or when computational efficiency is paramount, while SIM is more suitable for detailed QTL mapping where localization precision and power are critical.

By evaluating power, ROC performance, FDR, and robustness, we can make informed decisions about which method best suits their study design and data characteristics.

2.5 Data format

With the advent of high-throughput sequencing technologies, genetic data is now commonly available in formats such as Variant Call Format [6] (.vcf), which provides detailed information on genetic variations. This allows for more precise analyses, particularly when physical marker positions are known. Despite advances in marker-based approaches, the relative performance of SMT and SIM in specific experimental designs, such as backcross families, remains an area of active research.

2.5.1 Genotypes and marker positions

Marker Positions:

- Marker positions are crucial for QTL mapping, as they determine the physical and genetic distances between markers. These positions are defined by:
  - **Physical Position**: Measured in base pairs (bp) along the chromosome.
  - **Genetic Map Position**: Measured in centimorgans (cM) using a linkage map.
- Accurate marker positions enhance the resolution of QTL mapping, reducing positional ambiguity and improving the detection of linked QTLs.

Example of Marker Position Data:

Marker ID	Chromosome	Physical Position (bp)
M1	**1**	**1,000,000**
M2	**1**	**1,500,000**
M3	**2**	**2,000,000**

Source of Data:

- Marker and genotype data are often stored in Variant Call Format (VCF) files, which detail chromosomal positions and genotypic states.
- Genotype data is typically stored in **Variant Call Format (VCF)**, a widely used file format for describing genetic variants and detail chromosomal positions and genotypic states, Each entry in the VCF file specifies:
  - **CHROM**: Chromosome number.

- **POS**: Physical position of the marker in base pairs.
- **ID**: Marker identifier.
- **REF**: Reference allele.
- **ALT**: Alternate allele(s).
- **GT**: Genotype call for each individual.

VCF File Structure: A snippet of a VCF file might look like this:

```
#CHROM  POS      ID    REF ALT QUAL FILTER INFO FORMAT IND1 IND2 IND3 IND4 IND5
1       1000000  M1    A   G   .    .      .    GT     0/0  0/1  1/1  0/1  1/1
1       1500000  M2    T   C   .    .      .    GT     0/1  0/0  1/1  1/1  1/1
2       2000000  M3    G   A   .    .      .    GT     1/1  0/1  1/1  0/0  0/1
```

2.5.2 Trait values

Phenotypic Data:

- Quantitative traits are recorded for each individual in the population. These traits may include measurements such as:
  - Height, weight, yield, or other agricultural or biological metrics.
  - Disease resistance scores.
  - Biochemical or physiological measurements.

Data Structure:

- Trait values are stored in a tabular format, where rows represent individuals and columns represent traits. For example:

Individual	Trait 1 (Height)	Trait 2 (Weight)
1	150.2	60.1
2	145.5	58.7
…	…	…

- It is important to say that in our project we consider only a single trait case. Hence, each trait is dealt separately.

Normalization:

- Trait values are often normalized to meet the assumptions of statistical tests, such as linearity and normality. [2]

2.5.3 Output: Genetic map

Steps to Construct a Genetic Map:

1. Use marker genotype data and their physical positions.
2. Calculate recombination frequencies between markers using Haldane model [1] .
3. Convert recombination frequencies to map distances (cM).

Output Format:

The genetic map is typically presented as a table or visualized graphically, with each chromosome represented as a linear axis. Example:

Maker ID	Chromosome	Physical Position (bp)	Genetic Position (cM)
M1	1	1,000,000	5.2
M2	1	1,500,000	10.5
M3	2	2,000,000	15.8

Visualization:

- Genetic maps are often visualized as linear diagrams showing marker positions along chromosomes. These visualizations aid in interpreting results and planning downstream analyses.

2.5.4 Output: Estimated QTL(s) position and effect

Estimated QTL Position:

- QTL positions are reported as the most likely locations (physical or genetic) on the chromosome. For example:
    - **Chromosome 1, 15.2 cM**.

Effect Size:

- The effect size of a QTL indicates its contribution to the trait variation. This is reported as:

- **Additive Effect** (β): Measures the influence of the QTL on the trait per allele substitution.
- **Variance Explained** (R^2): Quantifies the proportion of total trait variation attributed to the QTL.

LOD Scores:

- **LOD (Logarithm of Odds) Scores** quantify the strength of the evidence for a QTL at a specific position. [21]

$$\text{LOD} = \log_{10}\left(\frac{\text{Likelihood of QTL model}}{\text{Likelihood of no QTL model}}\right)$$

- Higher LOD scores indicate stronger associations.

Output Format:

- QTL results are typically summarized in a table:

QTL ID	Chromosome	Position (cM)	Additive Effect(β)	Variance Explained (R^2)	LOD Score
QTL1	1	12.4	2.3	18%	5.1
QTL2	2	25.6	1.9	12%	4.3

Visualization:

- **QTL Mapping Plots**: Graphs showing LOD scores along chromosomes help visualize significant QTLs.
- Example: A Manhattan plot with chromosomes on the x-axis and LOD scores on the y-axis. [17]

2.6 Data simulation

Simulating data is a critical step in evaluating the performance of QTL mapping methodologies, such as Single Marker Test (SMT) and Single Interval Mapping (SIM). Simulation allows researchers to test the methods under controlled conditions with known QTL positions and traits, enabling precise assessment of their accuracy, statistical power, and computational efficiency. [22] [16]

Objectives of Data Simulation

1. **Controlled Testing**: To evaluate the performance of SMT and SIM in detecting QTLs under varying experimental conditions.
2. **Performance Metrics**: To assess accuracy, statistical power, and runtime efficiency by simulating traits with known genetic underpinnings.
3. **Robustness Testing**: To analyze the effects of data variations, such as marker density, sample size, and missing data, on method reliability.

Simulation Framework

1. Genotypic Data:
   - Simulated genotypes for markers and QTLs are generated based on known recombination rates and a defined genetic map.
   - The model assumes a single backcross family structure with M markers evenly distributed across the genome.
2. Phenotypic Data:
   - Quantitative traits are simulated using a Gaussian model:

$$T_i = \mu + \sum_q \mu_q \cdot (1\{g_{iq0} = a_q\} + 1\{g_{iq1} = a_q\}) + \epsilon_i$$

   where Ti represents the trait of individual i, μq is the effect size of QTL q, and ϵi~N(0, σ ^2) represents random noise. [15]

3. Genetic Map:
   - Recombination rates between adjacent markers are calculated using the Haldane mapping function, which assumes independent recombination events:

$$r_{m1m2} = 0.5 \cdot (1 - e^{-0.02 \cdot |x_{m2} - x_{m1}|})$$

   - Markers are then mapped to genetic positions, converting physical positions into centimorgans (cM). [12]

Simulation Scenarios:

- **Marker Density:** Varying numbers of markers per chromosome to test the effect of resolution on QTL detection.

- **Trait Heritability :** Simulating traits with different proportions of variance explained by genetic factors (e.g., low, medium, high heritability). **[23]**
- **Missing Data:** Introducing random missing genotype or phenotype data to evaluate robustness.
- **Noise Levels:** Adding controlled noise to phenotypic traits to mimic experimental errors.[5]

Implementation

The simulations will be implemented in the **R/qtl** [3] library within the **R** programming environment, which provides tools for:

- Generating synthetic genetic maps and marker data.
- Simulating quantitative traits based on defined QTL effects.
- Handling missing data and noise in simulated datasets.

Output and Evaluation

1. Siumlated Datasets:
    - Datasets will include marker genotypes, trait values, and known QTL positions.
    - Stored in a structured format suitable for downstream analysis (e.g., .vcf).
2. Performance Evaluation:
    - SMT and SIM will be applied to the simulated data.
    - Results will be compared using metrics such as LOD scores, false discovery rates, and computation time. [7]
    - Graphical outputs, such as Manhattan plots [17] , will visualize QTL detection across chromosomes.
3. Iterative Refinement:
    - Simulation parameters will be adjusted iteratively to mimic realistic biological scenarios and evaluate method reliability across a range of conditions.[16]

Conclusion

Simulated data provides a robust platform to evaluate QTL mapping methodologies under diverse scenarios. By systematically analyzing SMT and SIM, this study aims to identify their strengths and limitations, offering practical recommendations for genetic studies involving backcross populations.

3. Expected achievements

Focused Comparison of Methods

A targeted comparison of Single Marker Test (SMT) and Single Interval Mapping (SIM) specifically applied to a single backcross family, emphasizing their practical strengths and limitations in this context.

Specific Performance Metrics

Evaluation of both methods based on clearly defined performance indicators, such as:

- Accuracy in detecting QTLs.
- Statistical power for identifying marker-trait associations.
- Computational efficiency (runtime and resource usage).

Impact of Known Marker Positions

Analysis of how the known physical positions and spacing of markers influence the accuracy and resolution of SMT and SIM in identifying QTLs.

Context-Specific Recommendations

Actionable guidelines for selecting between SMT and SIM based on the unique characteristics of the dataset (e.g., marker density, experimental design), offering clear recommendations tailored to backcross family studies.

Simplified Workflow

Development of a practical and streamlined workflow for preprocessing .vcf files and implementing QTL mapping using SMT and SIM, providing a straightforward reference for similar studies.

Modest Contribution to Genetic Research

A focused contribution that enhances the understanding of SMT and SIM in QTL mapping, specifically within the scope of backcross families. The results will support researchers in applying these methods effectively for comparable datasets.

4. Requirements

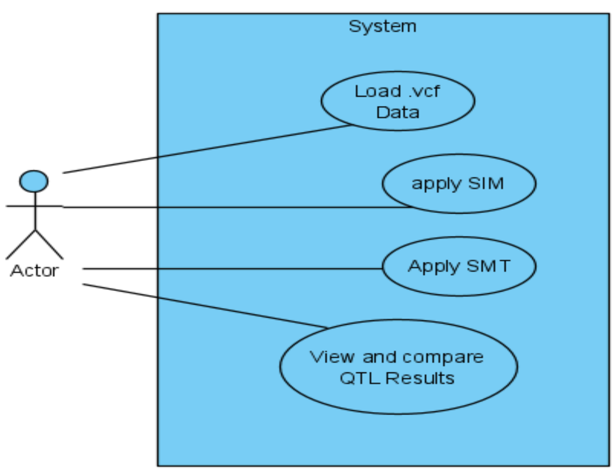
4.1 Functional Requirements (FRs)

1. **Genetic Data Input**:
   1.1) The software must support input in Variant Call Format (VCF), including genotypic and phenotypic data.
   1.2) It should process biallelic marker data and normalize trait values as needed.
2. **QTL Detection Algorithms**:
   2.1) Implement Single Marker Test (SMT) and Single Interval Mapping (SIM) for QTL detection.
3. **Genetic Map Construction**:
   3.1) Convert physical marker positions into genetic distances using the Haldane mapping function.
4. **Simulation Capabilities**:
   4.1) Enable the generation of synthetic genetic data, including markers and traits.
5. **Performance Metrics Output**:
   5.1) Generate metrics such as accuracy, LOD scores, statistical power, and false discovery rate for evaluation.
6. **Visualization Tools**:
   6.1) Provide graphical outputs like Manhattan plots to visualize QTL positions and LOD scores.
7. **Result Reporting**:
   7.1) Summarize detected QTL positions, marker effects, and statistical confidence.

4.2 Non-Functional Requirements (NFRs)

1. **Usability**:
   1.1) The interface should be intuitive and straightforward for genetic researchers with minimal training.
2. **Performance**:
   2.1) Ensure the software efficiently processes genetic data and executes QTL mapping methods.
3. **Scalability**:
   3.1) Support datasets with varying sizes.
4. **Reliability**:
   4.1) Achieve consistent results under different conditions, including variations in marker density and phenotypic noise.
5. **Maintainability**:
   5.1) The codebase should be modular and well-documented, facilitating updates and debugging.
6. **Compatibility**:
   6.1) Ensure compatibility with R/qtl library functions for both real and simulated data analysis.
7. **Statistical Validation**:
   7.1) Integrate permutation and resampling methods (e.g., bootstrap, jackknife) for robust statistical validation of QTL detections.

4.3 Use case diagram



5. Research Process

5.1 Data Preparation

- Real Data: Extract genetic marker and phenotypic data from Variant Call Format (VCF) files. Ensure only biallelic markers are retained and normalize trait values to satisfy QTL mapping assumptions.
- Simulated Data:
  - Generate datasets based on a single backcross family. Define parameters such as marker density, QTL effects, and noise levels.
  - Use Gaussian models for trait simulation, incorporating predefined QTL positions and heritability levels.

5.2 Genetic Map Construction

- Convert physical marker positions into genetic distances using the Haldane mapping function.
- Validate the genetic map for consistency with the genome structure.

5.3 Implementation of SMT and SIM

- Single Marker Test (SMT):
  - Perform linear regression for each marker independently to test for marker-trait associations.
  - Record significant markers based on p-values or thresholds.
- Single Interval Mapping (SIM):
  - Apply maximum likelihood estimation (MLE) to intervals between markers, considering flanking markers to estimate QTL positions.
  - Calculate LOD scores for each interval and identify significant QTLs.

5.4 Validation Methods

- Permutation Tests:
  - Shuffle phenotypic values to establish null distributions for statistical significance testing.

- Resampling Techniques:
    - Use bootstrapping to estimate confidence intervals and assess robustness of detected QTLs.

5.5 Visualizations

- Create Manhattan plots to visualize LOD scores across chromosomes.
- Summarize results in tables, including marker positions, detected QTLs, and statistical significance.

6. Evaluation plan

6.1 Evaluation Metrics

- **Accuracy:** Measure how closely detected QTL positions align with known positions in simulated datasets.
- **Statistical Power:** Quantify the ability to detect true QTLs and avoid false negatives.
- **False Discovery Rate (FDR):** Evaluate the proportion of false positives among detected QTLs.
- **LOD Scores:** Assess the statistical significance of QTL detections.
- **Computational Efficiency:** Compare runtime, memory usage, and scalability of SMT and SIM.

6.2 Testing Datasets

- **Simulated Data:**
    - Create synthetic datasets with varying conditions such as marker density, sample size, heritability, and noise levels.
- **Real Data:**
    - Use preprocessed VCF datasets to evaluate methods in a practical scenario.

6.3 Comparison Procedure

- Apply SMT and SIM to both simulated and real datasets.
- Analyze their performance based on the predefined evaluation metrics.
- Compare the simplicity of SMT versus the precision and computational requirements of SIM.

6.4 Statistical Validation

- Conduct permutation tests to determine LOD score significance thresholds.
- Use bootstrapping to evaluate consistency and robustness in QTL detection across replicates.
- Validate QTL effects by examining their additive contributions to phenotypic variance.

6.5 Visualization and Reporting

- Visualize findings through Manhattan plots and QTL effect size charts.
- Compile results into clear tables summarizing detected QTL positions, their effects, and statistical confidence.

6.6 Deliverables

- A thorough comparison of SMT and SIM, highlighting their trade-offs.
- Recommendations for selecting methods based on dataset characteristics.
- A final report summarizing the findings, supported by graphical and tabular outputs.

7. Summary

This capstone project explores the comparative performance of Single Marker Test (SMT) and Single Interval Mapping (SIM) for QTL detection in a single backcross family. SMT, characterized by its simplicity and computational efficiency, examines individual markers for their association with traits but lacks resolution for linked QTLs. SIM, on the other hand, uses likelihood-based approaches to enhance localization accuracy by incorporating information from flanking markers, albeit at higher computational costs.

The study involves:

1. Analyzing real VCF datasets and simulated data generated under controlled conditions.
2. Evaluating performance metrics such as accuracy, statistical power, false discovery rate (FDR), and computational efficiency.
3. Comparing results using both theoretical frameworks and practical experiments.

Simulated datasets enable precise performance assessment under varying conditions, such as marker density and trait heritability. Using the R/qtl library for analysis, the project provides robust methodologies for QTL mapping. The findings emphasize SIM's superior accuracy and power in complex scenarios and propose workflows for researchers to adopt based on dataset characteristics. The study contributes to advancing genetic mapping techniques and improving their application in agricultural and biological research.

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